Running Notes on the SCSI Algorithm

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Contents

1 Convergence of Gaussian w/ AWGN Channel

1.1 Setup

Consider a (true) data distribution π (in the paper it's $\pi = \gamma_{b,\Sigma}$) and an AWGN channel $F(x) = x + w$, where $w \sim \gamma$. Then the corrupted distribution becomes $\mu = \pi \star \gamma$ (in paper, $\mu = \gamma_{b,\Sigma+\mathbb{I}}$).

Define a linear interpolant

$$I_t = (1 - \sqrt{t})x + \sqrt{t}\, F(x) = x + \sqrt{t}\, w \tag{1.1}$$

where $x \sim \tilde{\pi}^{(k)}$ (your k'th guess for the true data distribution). Let $\mathrm{Law}(I_t) = \tilde{\pi}_t^{(k)}$, meaning $\tilde{\pi}_t^{(k)} = \tilde{\pi}^{(k)} \star \gamma_{t\mathbb{I}}$ (in paper, $\tilde{\pi}_t = \gamma_{\tilde{b},\tilde{\Sigma}+t\mathbb{I}}$). As a sanity check, the boundary conditions are $\tilde{\pi}_{t=0} = \tilde{\pi}^{(k)}$ is your guess for the data distribution, and $\tilde{\pi}_{t=1} = \tilde{\pi}^{(k)} \star \gamma$ is the corrupted distribution based on your guess for the data distribution. The $\tilde{\pi}^{(k)}$ satisfies the heat equation

$$\partial_t \tilde{\pi}_t^{(k)} = \frac{1}{2}\Delta \tilde{\pi}_t^{(k)} \tag{1.2}$$

Let $\tau = 1 - t$ and $q_\tau = \pi_{1-\tau}$. Plug q_τ into (1.2) this gives the reverse transport. In the SCSI algorithm, the dynamics for $q_\tau^{(k+1)}$ are due to the reverse drift (score) coming from the previous iteration $\nabla \log \tilde{\pi}_{1-\tau}^{(k)}$,

$$\begin{aligned}
\partial_\tau q_\tau^{(k+1)} &= -\frac{1}{2}\Delta q_\tau^{(k+1)} && (1.3)\\
&= -\frac{1+\epsilon_t}{2}\nabla \cdot (\nabla \log \tilde{\pi}_{1-\tau}^{(k)}\, q_\tau^{(k+1)}) + \frac{1}{2}\epsilon_t \Delta q_\tau^{(k+1)} && (1.4)
\end{aligned}$$

By Fokker-Planck, write the corresponding reverse SDE. Let $\text{Law}(Y_\tau^{(k+1)}) = q_\tau^{(k+1)}$, then

$$dY_\tau^{(k+1)} = \frac{1+\epsilon_\tau}{2}\nabla \log \tilde{\pi}_{1-\tau}^{(k)}(Y_\tau^{(k+1)})\, d\tau + \sqrt{\epsilon_\tau}\, dW_\tau \tag{1.5}$$

So to be very explicit

- Boundary conditions: $\text{Law}(Y_{\tau=0}^{(k+1)}) = \mu$ and $\text{Law}(Y_{\tau=1}^{(k+1)}) = \pi^{(k+1)}$
- Drift: $\nabla \log \tilde{\pi}_{1-\tau}^{(k)}$ satisfies (1.2) with boundary condition $\text{Law}(Y_{\tau=1}^{(k)}) = \tilde{\pi}_{t=0}^{(k)}$. In other words, $\tilde{\pi}_t^{(k)} = \text{Law}(Y_{\tau=1}^{(k)}) \star \gamma_{t\mathbb{I}}$.

Notation

- $t \in [0,1]$: Forward time, $t=0$ is clean data, $t=1$ is corrupted data.
- $\tau \in [0,1]$: Reverse time, $\tau = 1-t$.
- π: **True** (unobserved) clean data distribution
- μ: **True** (observed) corrupted data distribution.
- $\tilde{\pi}$ (or $\tilde{\pi}^{(k)}$): Algorithm's guess for the clean data at iteration k.
- $\text{Law}(Y_\tau) = q_\tau$.
- γ is an unit Gaussian, $\gamma_{b,\Sigma}$ is a Gaussian with mean b and covariance Σ.

1.2 Noise Scheduler

Assume the setup in the paper, except let's test how the choice of a noise-scheduler ϵ_t affects the convergence. Under constant noise scheduler, it was found that $\epsilon_t = 0$ gives the best performance, so let's see if different choices simplify to the same result.

The reverse SDE is

$$dY_\tau^{(k+1)} = A_\tau^{(k)}(Y_\tau - b^{(k)})d\tau + \sqrt{\epsilon_\tau}\, dW_\tau \tag{1.6}$$

where $A_\tau^{(k)} := -\frac{1+\epsilon_\tau}{2}(\Sigma^{(k)} + (1-\tau)\mathbb{I})^{-1}$. Via change-of-coordinates, the solution is generically given by

$$Y_\tau^{(k+1)} = b^{(k)} + \Phi^{(k)}(\tau,0)(Y_0^{(k+1)} - b^{(k)}) + \int_0^\tau \Phi^{(k)}(\tau,s)\sqrt{\epsilon_s}\, dW_s \tag{1.7}$$

where the solution map $\Phi(\tau,s)$ is given by

$$\Phi^{(k)}(\tau,s) = [\Phi^{(k)}(\tau)]^{-1}\Phi^{(k)}(s) \tag{1.8}$$

$$\Phi^{(k)}(\tau) = \exp\left(-\int_0^\tau A_s^{(k)}\, ds\right) \tag{1.9}$$

$$= \exp\left(\int_0^\tau \frac{1+\epsilon_s}{2}(\Sigma^{(k)} + (1-s)\mathbb{I})^{-1}\, ds\right) \tag{1.10}$$

To evaluate the integral, the trick is to change to a basis where $\Sigma^{(k)}$ is diagonal (which is always possible because $\Sigma^{(k)} \succ 0$). Additionally, the PSD causes $\Phi^{(k)}(\tau)$ to be symmetric.

Since everything in the generic solution (1.7) is Gaussian. That is $Y_0^{(k+1)} = \gamma_{b,\Sigma+\mathbb{I}}$, and $\int_0^\tau \Phi^{(k)}(\tau,s)\sqrt{\epsilon_s}\,dW_s \sim \mathcal{N}\left(0,\ \int_0^\tau \epsilon_s \Phi^{(k)}(\tau,s)\left(\Phi^{(k)}(\tau,s)\right)^T ds\right)$. We can use rules of Gaussian random variables to read off the the mean $b^{(k+1)}$ covariance $\Sigma^{(k+1)}$ of $Y_{\tau=1}^{(k+1)}$.

$$\boxed{\begin{aligned} b^{(k+1)} &= b^{(k)} + \Phi^{(k)}(1,0)\left(b - b^{(k)}\right) \\ \Sigma^{(k+1)} &= \Phi^{(k)}(1,0)\left(\Sigma + \mathbb{I}\right)\left(\Phi^{(k)}(1,0)\right)^T + \int_0^1 \epsilon_s \left(\Phi^{(k)}(1,s)\right)^2 ds \end{aligned}} \tag{1.11, 1.12}$$

1.2.1 Linear Noise

Let the noise scheduler be linear $\boxed{\epsilon_s = \alpha s}$ (s.t. $\alpha \geq 0$).

Compute Update The solution map becomes

$$\Phi^{(k)}(\tau) \sim \exp\left(\frac{1}{2}\int_0^\tau \frac{1+\alpha s}{\lambda^{(k)}+1-s}\,ds\right) \tag{1.13}$$

$$= e^{-\alpha\tau/2}\left(\frac{\lambda^{(k)}+1}{\lambda^{(k)}+1-\tau}\right)^{\frac{1}{2}+\frac{\alpha(\lambda^{(k)}+1)}{2}} \tag{1.14}$$

$$\sim e^{-\alpha\tau/2}\left[(\Sigma^{(k)}+\mathbb{I})(\Sigma^{(k)}+(1-\tau)\mathbb{I})^{-1}\right]^{\frac{1}{2}(\mathbb{I}+\alpha(\Sigma^{(k)}+\mathbb{I}))} \tag{1.15}$$

where $\sim$ denotes going back and forth, between the matrix & eigenvalue representations. As a sanity check, setting $\alpha = 0$ recovers the $\epsilon = 0$ for the constant noise scheduler in the original paper.

Additionally, some helpful formulas are

$$\Phi^{(k)}(1,s) = e^{\frac{\alpha}{2}(1-s)}\left[\Sigma^{(k)}(\Sigma^{(k)}+(1-s)\mathbb{I})^{-1}\right]^{\frac{1}{2}(\mathbb{I}+\alpha(\Sigma^{(k)}+\mathbb{I}))} \tag{1.16}$$

$$\Phi^{(k)}(1,0) = e^{\frac{\alpha}{2}}\left[\Sigma^{(k)}(\Sigma^{(k)}+\mathbb{I})^{-1}\right]^{\frac{1}{2}(\mathbb{I}+\alpha(\Sigma^{(k)}+\mathbb{I}))} \tag{1.17}$$

Now let's compute the integral

$$\mathcal{I} = \int_0^1 \underbrace{\epsilon_s}_{\alpha s}\left(\Phi^{(k)}(1,s)\right)^2 ds \tag{1.18}$$

We'll evaluate this in the basis which diagonalizes. Define the eigenvalues of $\left(\Phi^{(k)}(1,s)\right)^2$ as $V_k(s)$

$$V_k(s) = e^{\alpha(1-s)}\left(\frac{\lambda^{(k)}}{\lambda^{(k)}+1-s}\right)^{1+\alpha(\lambda^{(k)}+1)} \tag{1.19}$$

Because there's an exponential & multiplicative, there's hope of finding an identity relating $V_k(s)$ and $V_k'(s)$.

$$\log V_k(s) = \alpha(1-s) + (1+\alpha(\lambda^{(k)}+1))\left[\log\lambda^{(k)} - \log(\lambda^{(k)}+1-s)\right] \tag{1.20}$$

$$\frac{V_k'(s)}{V_k(s)} = \frac{d}{ds}\log V_k(s) \tag{1.21}$$

$$= -\alpha + \frac{1+\alpha(\lambda^{(k)}+1)}{\lambda^{(k)}+1-s} \tag{1.22}$$

$$= \frac{1+\alpha s}{\lambda^{(k)}+1-s} \tag{1.23}$$

Therefore we

$$\therefore\ \alpha s V_k(s) = (\lambda^{(k)} + 1 - s)V_k'(s) - V_k(s) \tag{1.24}$$

The original integral can be expressed (in diagonal basis)

$$\mathcal{I} \sim \int_0^1 \alpha s\, V_k(s)\, ds = \underbrace{\int_0^1 (\lambda^{(k)} + 1 - s)V_k'(s)\, ds}_{\mathcal{I}_1} - \int_0^1 V_k(s)\, ds \tag{1.25}$$

The first integral can be solved using integration by parts

$$\mathcal{I}_1 = (\lambda^{(k)} + 1 - s)V_k(s)\Big|_{s=0}^{1} - \int_0^1 (-1)V_k(s)\, ds \tag{1.26}$$

$$= \lambda^{(k)}V_k(1) - (\lambda^{(k)} + 1)V_k(0) + \int_0^1 V_k(s)\, ds \tag{1.27}$$

The $\int_0^1 V_k(s)\, ds$ cancels out, and we're left with

$$\mathcal{I} \sim \lambda^{(k)}V_k(1) - (\lambda^{(k)} + 1)V_k(0) \tag{1.28}$$

$$= \lambda^{(k)} - (\lambda^{(k)} + 1)V_k(0) \qquad V_k(1) = 1 \tag{1.29}$$

$$= \lambda^{(k)} - e^{\alpha}(\lambda^{(k)} + 1)\left(\frac{\lambda^{(k)}}{\lambda^{(k)} + 1}\right)^{1+\alpha(\lambda^{(k)}+1)} \tag{1.30}$$

$$\sim \Sigma^{(k)} - e^{\alpha}(\Sigma^{(k)} + \mathbb{I})[\Sigma^{(k)}(\Sigma^{(k)} + \mathbb{I})^{-1}]^{\mathbb{I}+\alpha(\Sigma^{(k)}+\mathbb{I})} \tag{1.31}$$

$$= \Sigma^{(k)} - (\Sigma^{(k)} + \mathbb{I})\left(\Phi^{(k)}(1,0)\right)^2 \tag{1.32}$$

Plugging this into the covariance update (1.12), we find

$$\Sigma^{(k+1)} = \Phi^{(k)}(1,0)(\Sigma + \mathbb{I})\left(\Phi^{(k)}(1,0)\right)^T + \Sigma^{(k)} - (\Sigma^{(k)} + \mathbb{I})\left(\Phi^{(k)}(1,0)\right)^2 \tag{1.33}$$

$$\boxed{\Sigma^{(k+1)} = \Sigma^{(k)} + \Phi^{(k)}(1,0)\ \left(\Sigma - \Sigma^{(k)}\right)\Phi^{(k)}(1,0)} \tag{1.34}$$

I've also used the fact that $\Phi^{(k)}(1,0)$ commutes with $\Sigma^{(k)}$. Notice this is the exact same dynamics as the case study in the paper, just relabel $\Phi^{(k)}(1,0) \mapsto B_k$. From here we can bound the convergence.

Convergence Make the substitution $E^{(k)} := \Sigma - \Sigma^{(k)}$, then (1.34) becomes

$$E^{(k+1)} = E^{(k)} - \Phi^{(k)}(1,0)\, E^{(k)}\, \Phi^{(k)}(1,0) \tag{1.35}$$

I want to bound the spectral norm $\|E^{(k+1)}\|$.

Fact 1 *Let X be a symmetric matrix, and B be a symmetric matrix s.t. $0 \prec B \prec \mathbb{I}$*

$$\|X - BXB\| \leq (1 - \lambda_{min}(B)^2)\|X\| \tag{1.36}$$

For proof, see original paper.

Additionally, for all k

$$\lambda_{min}(\Sigma_k) \geq \min(\lambda_{min}(\Sigma), \lambda_{min}(\Sigma_0)) =: \eta \tag{1.37}$$

which follows from the same argument in the original paper. Now to finish, I just need to compute $\lambda_{min}(\Phi^{(k)}(1,0))$

$$\lambda_{min}(\Phi^{(k)}(1,0)) = e^{\alpha/2}\left(\frac{\lambda_{min}(\Sigma^{(k)})}{\lambda_{min}(\Sigma^{(k)})+1}\right)^{\frac{1}{2}[1+\alpha(\lambda_{min}(\Sigma^{(k)})+1)]} \geq e^{\alpha/2}\left(\frac{\eta}{\eta+1}\right)^{\frac{1}{2}[1+\alpha(\eta+1)]} \tag{1.38}$$

where we used it the fact it is monotonically increasing in $\lambda_{min}(\Sigma^{(k)})$.

Proposition 1 (Gaussian AWGN with Linear Noise Scheduler, Convergence) *Let* $\eta := \lambda_{min}(\Sigma_k) \geq \min(\lambda_{min}(\Sigma), \lambda_{min}(\Sigma_0))$ *and assume* $\eta > 0$*. Then*

$$\|\Sigma - \Sigma_k\| \leq \left(1 - e^{\alpha}\left(\frac{\eta}{\eta+1}\right)^{1+\alpha(\eta+1)}\right)^k \|\Sigma - \Sigma_0\| \tag{1.39}$$

$$\|b - b_k\|^2 \leq \left(1 - e^{\alpha}\left(\frac{\eta}{\eta+1}\right)^{1+\alpha(\eta+1)}\right)^k \|b - b_0\|^2 \tag{1.40}$$

We find optimal convergence when $\alpha = 0$*, which recovers the ODE solution.*

1.3 Arbitrary Interpolant & Noise Schedulers

Consider the interpolant

$$I_t = (1-\sqrt{\alpha_t})x + \sqrt{\alpha_t}\,\mathcal{F}(x) = x + \sqrt{\alpha_t}\,w \tag{1.41}$$

where α_t is monotonically increasing w/ boundary conditions $\alpha_0 = 0$ and $\alpha_1 = 1$. Therefore $\tilde{\pi}_t = \pi \star \gamma_{\alpha_t \mathbb{I}}$, which satisfies a modified Heat Semigroup

$$\partial_t \tilde{\pi}_t = \frac{\dot{\alpha}_t}{2}\Delta \tilde{\pi}_t \tag{1.42}$$

The reverse PDE becomes

$$\begin{aligned}
\partial_t q_\tau &= -\frac{\dot{\alpha}_\tau}{2}\Delta q_\tau && (1.43)\\
&= -\frac{\dot{\alpha}_\tau}{2}\nabla\cdot(\nabla \log q_\tau\, q_\tau) && (1.44)\\
&= -\frac{\dot{\alpha}_\tau + \epsilon_\tau}{2}\nabla\cdot(\nabla \log q_\tau\, q_\tau) + \frac{\epsilon_\tau}{2}\Delta q_\tau && (1.45)
\end{aligned}$$

Which has the reverse SDE

$$\begin{aligned}
dY_\tau^{(k+1)} &= \frac{\dot{\alpha}_\tau + \epsilon_\tau}{2}\nabla \log \tilde{\pi}_{1-\tau}^{(k)}(Y_\tau^{(k+1)})\,d\tau + \sqrt{\epsilon_\tau}\,dW_\tau && (1.46)\\
\nabla \log \tilde{\pi}_{1-\tau}^{(k)}(x) &= -[\Sigma^{(k)} + (1-\alpha_\tau)\mathbb{I}]^{-1}(x - b^{(k)}) && (1.47)\\
\therefore Y_\tau^{(k+1)} &= b^{(k)} + \Phi^{(k)}(\tau,0)(Y_0^{(k)} - b^{(k)}) + \int_0^\tau \Phi^{(k)}(\tau,s)\sqrt{\epsilon_s}dW_s && (1.48)
\end{aligned}$$

Let $\Phi^{(k)}(\tau,s) := \Phi^{(k)}(\tau)^{-1}\Phi^{(k)}(s)$.

$$\Phi(\tau) = \exp\left(\int_0^\tau \underbrace{\frac{\dot{\alpha}_\tau + \epsilon_\tau}{2}[\Sigma^{(k)} + (1-\alpha_s)\mathbb{I}]^{-1})}_{:=A_s}\,ds\right) \tag{1.49}$$

Question: Can the covariance of $\int_0^1 \Phi^{(k)}(\tau, s)\sqrt{\epsilon_s}\, dW_s$ be generically written as $\Sigma^{(k)} - (\Sigma^{(k)} + \mathbb{I})\Phi^{(k)}(1,0)^2$? That is you don't specify ϵ_s, α_s. The answer in the previous section seems to indicate this...
Answer:

$$\mathcal{I} := \mathrm{Cov}\left[\int_0^1 \Phi^{(k)}(1,s)\sqrt{\epsilon_s}\, dW_s\right] \tag{1.50}$$
$$= \int_0^1 \Phi^{(k)}(1,s)^2\, \epsilon_s\, ds \tag{1.51}$$

Due to A_s's definition, $\epsilon_s\mathbb{I} = 2A_s[\Sigma^{(k)} + (1-\alpha_s)\mathbb{I}] - \dot{\alpha}_s\mathbb{I}$

$$\mathcal{I} = \underbrace{\int_0^1 2\, A_s\, [\Sigma^{(k)} - (1-\alpha_s)\mathbb{I}]\, \Phi^{(k)}(1,s)^2\, ds}_{\mathcal{I}_1} - \int \dot{\alpha}_s \Phi^{(k)}(1,s)^2\, ds \tag{1.52}$$

Integrate by parts $\int U\, dV = UV - \int V\, dU$. Use the substitution

- $U = \Sigma^{(k)} + (1-\alpha_s)\mathbb{I} \implies dU = -\dot{\alpha}_s\mathbb{I}\, ds$
- $V = \Phi^{(k)}(1,s)^2 \implies dV = 2\, A_s\, \Phi^{(k)}(1,s)^2\, ds$

$$\mathcal{I}_1 = [\Sigma^{(k)} + (1-\alpha_s)\mathbb{I}]\Phi^{(k)}(1,s)^2\Big|_{s=0}^{1} - \int \Phi^{(k)}(1,s)^2(-\dot{\alpha}_s)\, ds \tag{1.53}$$

Notice the last part cancels out when plugged back into (1.52).

$$\mathcal{I} = \left(\Sigma^{(k)}(\Phi^{(k)}(1,1)^2 - \Phi^{(k)}(1,0))^2\right) + \left((1-\alpha_1)\Phi^{(k)}(1,1)^2 - (1-\alpha_0)\Phi^{(k)}(1,0)^2\right) \tag{1.54}$$

Recall $\alpha_0 = 0$ and $\alpha_1 = 1$, and notice $\Phi^{(k)}(\tau,\tau) = \mathbb{I}$.

$$\mathcal{I} = \Sigma^{(k)} - (\Sigma^{(k)} + \mathbb{I})\Phi^{(k)}(1,0)^2 \tag{1.55}$$

■

Question: What is update scheme for arbitrary α_t, ϵ_t?
Answer:

$$b^{(k+1)} = b^{(k)} + \Phi^{(k)}(1,0)(b - b^{(k)}) \tag{1.56}$$
$$\Sigma^{(k+1)} = \Sigma^{(k)} + \Phi^{(k)}(1,0)(\Sigma - \Sigma^{(k)})\Phi^{(k)}(1,0) \tag{1.57}$$

where

$$\Phi(1,0) = \exp\left(-\int_0^1 \frac{\dot{\alpha}_\tau + \epsilon_\tau}{2}[\Sigma^{(k)} + (1-\alpha_s)\mathbb{I}]^{-1})\, ds\right) \tag{1.58}$$

2 MRA Channel

A natural question is to ask how do we do the SCSI framework when working on problems that have symmetries. For example:

Definition 1 (Multi Reference Alignment (MRA)) *Consider an image $x \in \mathbb{R}^d$. The MRA channel randomly shifts & blurs it*

$$\mathcal{F}(x) = g \cdot x + w \tag{2.1}$$

where $w \sim \mathcal{N}(0, \sigma^2 \mathbb{I})$ and $g \sim G$ (where G is Haar measure over the translation group with periodic boundary conditions).

Difficulty with learning & lifting as a solution Consider having access to clean sample $x \sim \pi$, and want to make a generative model over $\mu = \mathcal{K}\pi$. I'll choose to solve this using the linear interpolant

$$I_t = (1-t)x + t\ \mathcal{F}(x), \qquad \mathcal{F}(x) = g.x + w \tag{2.2}$$

The optimal drift becomes $b_t(y) = \mathbb{E}[\dot{I}_t | I_t = y] = \mathbb{E}[g.x + w - x | I_t = y]$. The inside quantity is a very weird image. To illustrated this, assume π is the empirical distribution over MNIST images; the $g.x - x$ is the digit plus a negative of the same digit moved, then add noise. It seems this quantity will be extremely difficult to learn.

Instead, what if we solved the lifted problem

$$I_t | x = (1-t)\, z + t\, \mathcal{F}(x) \tag{2.3}$$

where the optimal drift is $b_t(y|x) = \mathbb{E}[\dot{I}_t | I_t = y, x]$. This drift is so nice that you can actually analytically sovle it

2.1 Access to Clean Data, MRA Corruptions

2.1.1 Solution to Optimal Drift Field

Consider the stochastic interpolant

$$I_t = \alpha_t\, z + \beta_t\, \mathcal{F}(x) = \alpha_t\, z + \beta_t (g \cdot x + w) \tag{2.4}$$

where $z \sim \mathcal{N}(0, \mathbb{I})$, g is sampled uniformly from the translation group (w/ periodic boundary conditions), $x \sim \pi$ (let's assume it's MNIST images), and $w \sim \mathcal{N}(0, \sigma^2 \mathbb{I})$.

I want to compute the optimal drift

$$\begin{aligned} b_t(y|x) &= \mathbb{E}[\dot{I}_t | I_t = y, x] && (2.5)\\ &= \underbrace{\mathbb{E}[\dot{\alpha}_t z + \dot{\beta}_t w | I_t = y, x]}_{\equiv \mathcal{I}_1} + \dot{\beta}_t \underbrace{\mathbb{E}[g \cdot x | I_t = y, x]}_{\equiv \mathcal{I}_2(y,x)} && (2.6) \end{aligned}$$

Let's first compute $\mathcal{I}_2$.

$$\begin{aligned} \mathcal{I}_2(y,x) &= \mathbb{E}[g \cdot x | I_t = y, x] && (2.7)\\ &= \mathbb{E}_{g|y,x}[\mathbb{E}[g \cdot x | I_t = y, x, g]] && (2.8)\\ &= \int p(g|y,x)\, g \cdot x\, dg && (2.9)\\ &= \int \frac{p(y|g,x) p(g|x)}{\int p(y|g',x) p(g'|x)\, dg'} (g \cdot x)\, dg && (2.10) \end{aligned}$$

Notice that $y = \alpha_t z + \beta_t(g \cdot x + w)$, so $p(y|g,x) = \mathcal{N}(y; \beta_t\, g \cdot x,\ s_t^2 \mathbb{I})$ where $s_t^2 = \alpha_t^2 + \beta_t^2 \sigma^2$. Additionally $p(g|x) = \text{Uniform}(G)$ (which is translation w/ periodic boundary condtions). So $\int f(g) p(g|x)\, dg = \mathbb{E}_g[f(g)] = \frac{1}{|G|} \sum_{g \in G} f(g)$ is the sum over translations

$$p(g|y,x) = \frac{\exp\left(-\frac{\|y - \beta_t\, g \cdot x\|^2}{2s_t^2}\right)}{\sum_{g' \in G} \exp\left(-\frac{\|y - \beta_t\, g' \cdot x\|^2}{2s_t^2}\right)} \tag{2.11}$$

Use $\|y - \beta_t g \cdot x\|^2 = \|y\|^2 + \beta_t^2 \|g \cdot x\|^2 - 2\beta_t \langle y, g \cdot x \rangle$. Notice vector magnitude is invariant under permutations of entries, so $\|g \cdot x\| = \|x\|$.

$$p(g|y,x) = \frac{\exp\left(\frac{\beta_t}{s_t^2} \langle y, g \cdot x \rangle\right)}{\sum_{g' \in G} \exp\left(\frac{\beta_t}{s_t^2} \langle y, g' \cdot x \rangle\right)} =: \text{softmax}_g \left(\frac{\beta_t}{s_t^2} \langle y, g \cdot x \rangle\right) \tag{2.12}$$

So now we can pug this into $\mathcal{I}_2$

$$\mathcal{I}_2 = \sum_{g \in G} (g \cdot x)\, p(g|y,x) \tag{2.13}$$

Now let's compute $\mathcal{I}_1$.

$$\begin{aligned} \mathcal{I}_1 &= \mathbb{E}[\dot{\alpha}_t z + \dot{\beta}_t w | I_t = y, x] && (2.14) \\ &= \mathbb{E}_g[\mathbb{E}[\dot{\alpha}_t z + \dot{\beta}_t w | I_t = y, x, g] | y, x] && (2.15) \end{aligned}$$

From here notice that conditioned on x, g the random variables $U := \dot{\alpha}_t z + \dot{\beta}_t w$ and $V := I_t$ are jointly Gaussian. This is because I_t is an affine combination of z, w (which are jointly Gaussian).

Fact 2 *Let (U, V) be jointly Gaussian. Then the conditional expectation value becomes*

$$\mathbb{E}[U|V = v] = \mu_U + \Sigma_{UV} \Sigma_{VV}^{-1} (v - \mu_V) \tag{2.16}$$

where $\mu_U = \mathbb{E}[U]$, $\mu_V = \mathbb{E}[V]$, and $\Sigma_{UV} = Cov(U, V)$. This implies $\Sigma_{VV} = Var(V)$.

Proof: See StackExchange.

In our case,

- $\mu_U = \mathbb{E}[\dot{\alpha} z + \dot{\beta}_t w | x, g] = 0$
- $\mu_V = \mathbb{E}[\alpha_t z + \beta_t(g \cdot x + w) | x, g] = \beta_t\, g \cdot x$
- $\Sigma_{UV} = \text{Cov}(U, V | g, x) = (\dot{\alpha}_t \alpha_t + \dot{\beta}_t \beta_t \sigma^2) \mathbb{I}$
- $\Sigma_{VV} = \text{Var}(V | g, x) = s_t^2 \mathbb{I}$

Therefore $\mathcal{I}_1$ becomes

$$\begin{aligned} \mathcal{I}_1 &= \frac{\dot{\alpha}_t \alpha_t + \dot{\beta}_t \beta_t \sigma^2}{s_t^2} \mathbb{E}_g \left[y - \beta_t\, g \cdot x | I_t = y, x\right] && (2.17) \\ &= \frac{\dot{\alpha}_t \alpha_t + \dot{\beta}_t \beta_t \sigma^2}{s_t^2} (y - \beta_t \mathcal{I}_2) && (2.18) \end{aligned}$$

Plugging it back into (2.6) yields the exact drift

$$\boxed{b_t(y|x) = \mu_t\, y + \lambda_t \sum_{g\in G} (g \cdot x)\, p(g|y, x)} \tag{2.19}$$

where $\mu_t := (\dot{\alpha}_t \alpha_t + \dot{\beta}_t \beta_t \sigma^2)/s_t^2$, $\lambda_t = \dot{\beta}_t - \mu_t \beta_t$, $p(g|y, x) = \text{softmax}_g \left(\frac{\beta_t}{s_t^2} \langle y, g \cdot x\rangle\right)$, and $s_t^2 = \alpha_t^2 + \beta_t^2 \sigma^2$.

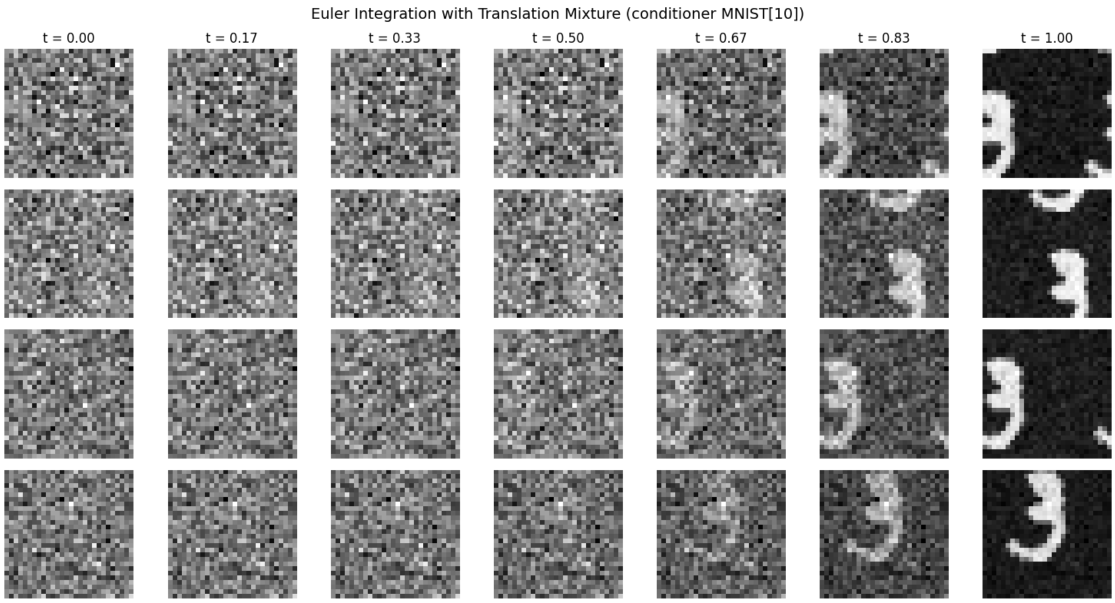

Figure 1: Numerically integrates $dX_t = b_t(X_t|x)\, dt$, where $x \sim$ MNIST, and boundary condition $X_0 \sim N(0, \mathbb{I})$. The drift is the analytic drift found in (2.19). In this case, the conditioner x is the digit 3.

Comment on Softmax The softmax structure comes from the group of translations (on lattice w/ periodic boudary conditions) being finite. If we redid this with $G = \text{SO}(3)$ rotations, then the drift field would be

$$b_t(y|x) = \mu_t\, y + \lambda_t \frac{\mathbb{E}_{g\sim \text{SO}(3)} \left[(g \cdot x) \exp(\frac{\beta_t}{s_t^2} \langle y, g \cdot x\rangle)\right]}{\mathbb{E}_{g\sim \text{SO}(3)}[\exp(\frac{\beta_t}{s_t^2} \langle y, g \cdot x\rangle)]} \tag{2.20}$$

2.1.2 Comments on Neural Network Architecture

Because we know the optimal solution, perhaps we can gleme some insights into how the neural network architecture should look like.

- The drift field is jointly equivariant $b_t(h \cdot y|h \cdot x) = h \cdot b_t(y|x)$, where $h \in G$ (translation group).

  Proof: Let $h \in G$ (translation group)

$$b_t(y|h \cdot x) = \mu_t y + \lambda_t \sum_{g\in G} (g \cdot x) p(g|y, h \cdot x) \tag{2.21}$$

$$p(g|y, h \cdot x) = \frac{\exp\left(\frac{\beta_t}{s_t^2} \langle y, g \cdot h \cdot x\rangle\right)}{\sum_{g'\in G} \exp\left(\frac{\beta_t}{s_t^2} \langle y, g' \cdot h \cdot x\rangle\right)} \tag{2.22}$$

Notice that $\langle y, g \cdot h \cdot x \rangle = \langle h^{-1} \cdot y, g \cdot x \rangle$, that is translating x forward is the same as had you translated y backwards.

$$\therefore p(g|y, h \cdot x) = p(g|h^{-1} \cdot y, x) \implies p(g|h \cdot y, h \cdot x) = p(g|y, x) \tag{2.23}$$

So the drift field becomes

$$\begin{align} b_t(h \cdot y|h \cdot x) &= \mu_t(h \cdot y) + \lambda_t \sum_{g \in G} (g \cdot h \cdot x) p(g|h \cdot y, h \cdot x) \tag{2.24} \\ &= h \cdot \Big(\mu_t y + \lambda_t \cdot \sum_{g \in G} (g \cdot x) p(g|y, x) \Big) \tag{2.25} \\ &= h \cdot b_t(y|x) \tag{2.26} \end{align}$$

■

2.2 Access to Corrupted Data, MRA Corruptions

Now consider the inverse problem. You have access to $Y \sim \nu$ (corrupted data distribution). This was created by running a clean piece of data $X \sim \pi$ through a corruption channel $\mathcal{F}(X) = G \cdot X + W = y$, where G are sampled uniformaly from translation group w/ periodic boundary conditions, and $W \sim \mathcal{N}(0, \sigma^2 \mathbb{I})$. We assume no model mispecification, so $\mathcal{F}(X) = Y \sim \nu$.

2.2.1 Solution to Optimal Drift Field, Gaussian

Setup Consider the optimal interpolant from noise to the clean data (up to translation)

$$I_t = \alpha_t Z + \beta_t \, G \cdot X \tag{2.27}$$

where $Z \sim \mathcal{N}(0, \mathbb{I})$ and $X \sim \pi$. In practice, we only have access to corrupted data Y, so I want to learn the optimal drift field, which flows to $X \sim \pi$, but is conditioned on $Y \sim \nu$.

$$b_t(x|y) = \mathbb{E}[\dot{I}_t | I_t = x | Y = y] \tag{2.28}$$

To make an attempt at solving this analytically, let's assume the clean data is $\pi = \mathcal{N}(\mu, \Sigma)$. And the translation acts on a 1D periodic lattice (chain), so it's a permutation matrix

$$G \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{pmatrix} = \begin{pmatrix} x_d \\ x_1 \\ \vdots \\ x_{d-1} \end{pmatrix} \tag{2.29}$$

Optimal Drift Field Calulation

$$\begin{align} b_t(x|y) &= \mathbb{E}[\dot{I}_t | I_t = x, Y = y] \tag{2.30} \\ &= \mathbb{E}_{g|x,y} \mathbb{E}[\dot{I}_t | I_t = x, Y = y, G = g] \tag{2.31} \\ &= \dot{\alpha}_t \, \mathbb{E}_{g|x,y} \mathbb{E}[Z | I_t = x, Y = y, G = g] + \dot{\beta}_t \, \mathbb{E}_{g|x,y} \mathbb{E}[G \cdot X | I_t = x, Y = y, G = g] \tag{2.32} \end{align}$$

By (2.27), we know $Z = \frac{1}{\alpha_t}(I_t - \beta_t\, G \cdot X)$. And by

$$= \frac{\dot\alpha_t}{\alpha_t}(x - \beta_t \mathbb{E}_{g|x,y}\, \mathbb{E}[G \cdot X|...]) + \dot\beta_t \mathbb{E}_{g|x,y} \mathbb{E}[G \cdot X|...] \tag{2.33}$$

$$= \frac{\dot\alpha_t}{\alpha_t} x + \left(\dot\beta_t - \frac{\dot\alpha_t \beta_t}{\alpha_t}\right) \mathbb{E}_{g|x,y}\, \mathbb{E}[G \cdot X | I_t = x, Y = y, G = g] \tag{2.34}$$

Notice the expectation value is over the discrete translation group

$$\mathbb{E}_{g|x,y}[f(g,x,y)] = \sum_{g \in G} p(g|I_t = x, Y = y) f(g,x,y) \tag{2.35}$$

First let's compute the probability $p(g|I_t = x, Y = y)$; by Bayes rule

$$p(g|I_t = x, Y = y) = \frac{p(x,y|G = g)p(G = g)}{\sum_{g' \in \mathcal{G}} p(x,y|G = g')P(G = g')} \tag{2.36}$$

Since $p(G) = 1/|\mathcal{G}|$ (sampled uniformly), it cancels out.

We are left trying to determine the joint $I_t, Y|G = g$.

(1) First let's inspect $Y|G = g$.

$$Y|(G = g) = \mathcal{F}(X) = g \cdot X + W \tag{2.37}$$

So Law$(Y|G = g)$, is an instance of a permuted Gaussian $\pi = \gamma_{\mu_g, \Sigma_g}$ convolved with another gaussian $\gamma_{0,\sigma^2 \mathbb{I}}$, resulting in Law$(Y|G = g) = \gamma_{\mu_g,\, \Sigma_g + \sigma^2 \mathbb{I}}$. The permuted mean & covariance are given by $\mu_g = g\mu$ and $\Sigma_g = g \Sigma g^T$

(2) Now let's inspect $I_t|G = g$

$$I_t|(G = g) = \alpha_t Z + \beta_t\, g \cdot X \tag{2.38}$$

So once again it's a Gaussian convolved with another Gaussian, Law$(I_t|G = g) = \gamma_{\beta_t \mu_g,\, \beta_t^2 \Sigma_g + \alpha_t^2 \mathbb{I}}$.

(3) We can now find the correlation

$$\mathrm{Cov}(I_t, Y|G = g) = \mathrm{Cov}(\alpha_t Z + \beta_t\, g \cdot X, g \cdot X + W|G = g) \tag{2.39}$$

$$= \alpha_t\, \mathrm{Cov}(Z, g \cdot X) + \beta_t \mathrm{Var}(g \cdot X) + \alpha_t \mathrm{Cov}(Z, W) + \beta_t \mathrm{Cov}(g \cdot X, W) \tag{2.40}$$

$$= \beta_t \mathrm{Var}(g \cdot X) \tag{2.41}$$

$$= \beta_t \Sigma_g \tag{2.42}$$

(4) Putting it all together, we find

$$\begin{pmatrix} I_t \\ Y \end{pmatrix} \Big| G = g \sim \mathcal{N}\left(\begin{pmatrix} \beta_t \mu_g \\ \mu_g \end{pmatrix}, \begin{pmatrix} \beta_t^2 \Sigma_g + \alpha_t^2 \mathbb{I} & \beta_t \Sigma_g \\ \beta_t \Sigma_g & \Sigma_g + \sigma^2 \mathbb{I} \end{pmatrix} \right) \tag{2.43}$$

where I'm using block matrix notation.

This leaves us with

$$p(g|I_t = x, Y = y) = \frac{\mathcal{N}\left(\begin{pmatrix} x \\ y \end{pmatrix}; \begin{pmatrix} \beta_t \mu_g \\ \mu_g \end{pmatrix}, \begin{pmatrix} \beta_t^2 \Sigma_g + \alpha_t^2 \mathbb{I} & \beta_t \Sigma_g \\ \beta_t \Sigma_g & \Sigma_g + \sigma^2 \mathbb{I} \end{pmatrix}\right)}{\sum_{g' \in \mathcal{G}} \mathcal{N}\left(\begin{pmatrix} x \\ y \end{pmatrix}; \begin{pmatrix} \beta_t \mu_{g'} \\ \mu_{g'} \end{pmatrix}, \begin{pmatrix} \beta_t^2 \Sigma_{g'} + \alpha_t^2 \mathbb{I} & \beta_t \Sigma_{g'} \\ \beta_t \Sigma_{g'} & \Sigma_{g'} + \sigma^2 \mathbb{I} \end{pmatrix}\right)} \tag{2.44}$$

Now let's calculate the inside conditional expectation

$$\mathbb{E}[G \cdot X | I_t = x, Y = y, G = g] \tag{2.45}$$

Let $\tilde{X} := G \cdot X$, meaning $\tilde{X}|G = g \sim \mathcal{N}(\mu_g, \Sigma_g)$. There are two (conditionally) independent observations of $\tilde{X}$,

$$Y = \tilde{X} + W \quad \implies p(y|\tilde{x}) = \mathcal{N}(y;\ \tilde{x}, \sigma^2 \mathbb{I}) \tag{2.46}$$

$$I_t = \beta_t \tilde{X} + \beta_t Z \quad \implies p(x|\tilde{x}) \sim \mathcal{N}(x;\ \beta_t \tilde{x}, \alpha_t^2 \mathbb{I}) \tag{2.47}$$

So by Bayes theorem, we can write the posterior

$$p(\tilde{x}|x, y, g) = \frac{p(x, y|\tilde{x}, g) p(\tilde{x}|g)}{Z} \tag{2.48}$$

$$= \frac{p(x|\tilde{x}, g) p(y|\tilde{x}, g) p(\tilde{x}|g)}{Z} \tag{2.49}$$

Gaussian times Gaussian (in probability density functions) is Gaussian, implying the posterior will be a Gaussian. A tedious calculation shows [Gemini says this, check later.]

$$p(\tilde{x}|x, y, g) = \mathcal{N}(\tilde{x};\ \mu_{post}, \Sigma_{post}) \tag{2.50}$$

$$\Sigma_{post} = \left(\Sigma_g^{-1} + \left(\tfrac{1}{\sigma^2} + \tfrac{\beta_t^2}{\alpha_t^2}\right) \mathbb{I}\right)^{-1} \tag{2.51}$$

$$\mu_{post} = \Sigma_{post} \left(\Sigma_g^{-1} \mu_g + \tfrac{1}{\sigma^2} y + \tfrac{\beta_t}{\alpha_t^2} x\right) \tag{2.52}$$

This means the conditional expectation is

$$\mathbb{E}[G \cdot X | I_t = x, Y = y, G = g] = \left(\Sigma_g^{-1} + \left(\tfrac{1}{\sigma^2} + \tfrac{\beta_t^2}{\alpha_t^2}\right) \mathbb{I}\right)^{-1} \left(\Sigma_g^{-1} \mu_g + \tfrac{1}{\sigma^2} y + \tfrac{\beta_t}{\alpha_t^2} x\right) \tag{2.53}$$

Putting this all together, we have an analytic expression for the drift

$$\boxed{b_t(x|y) = \frac{\dot{\alpha}_t}{\alpha_t} x + \left(\dot{\beta}_t + \frac{\dot{\alpha}_t}{\alpha_t} \beta_t\right) \sum_{g \in \mathcal{G}} p(g|I_t = x, Y = y)\, m_{g,t}(x, y)} \tag{2.54}$$

where

$$p(g|I_t = x, Y = y) = \frac{\mathcal{N}\left(\begin{pmatrix} x \\ y \end{pmatrix}; \begin{pmatrix} \beta_t \mu_g \\ \mu_g \end{pmatrix}, \begin{pmatrix} \beta_t^2 \Sigma_g + \alpha_t^2 \mathbb{I} & \beta_t \Sigma_g \\ \beta_t \Sigma_g & \Sigma_g + \sigma^2 \mathbb{I} \end{pmatrix}\right)}{\sum_{g' \in \mathcal{G}} \mathcal{N}\left(\begin{pmatrix} x \\ y \end{pmatrix}; \begin{pmatrix} \beta_t \mu_{g'} \\ \mu_{g'} \end{pmatrix}, \begin{pmatrix} \beta_t^2 \Sigma_{g'} + \alpha_t^2 \mathbb{I} & \beta_t \Sigma_{g'} \\ \beta_t \Sigma_{g'} & \Sigma_{g'} + \sigma^2 \mathbb{I} \end{pmatrix}\right)} \tag{2.55}$$

$$m_{g,t}(x, y) = \left(\Sigma_g^{-1} + \left(\tfrac{1}{\sigma^2} + \tfrac{\beta_t^2}{\alpha_t^2}\right) \mathbb{I}\right)^{-1} \left(\Sigma_g^{-1} \mu_g + \tfrac{1}{\sigma^2} y + \tfrac{\beta_t}{\alpha_t^2} x\right) \tag{2.56}$$

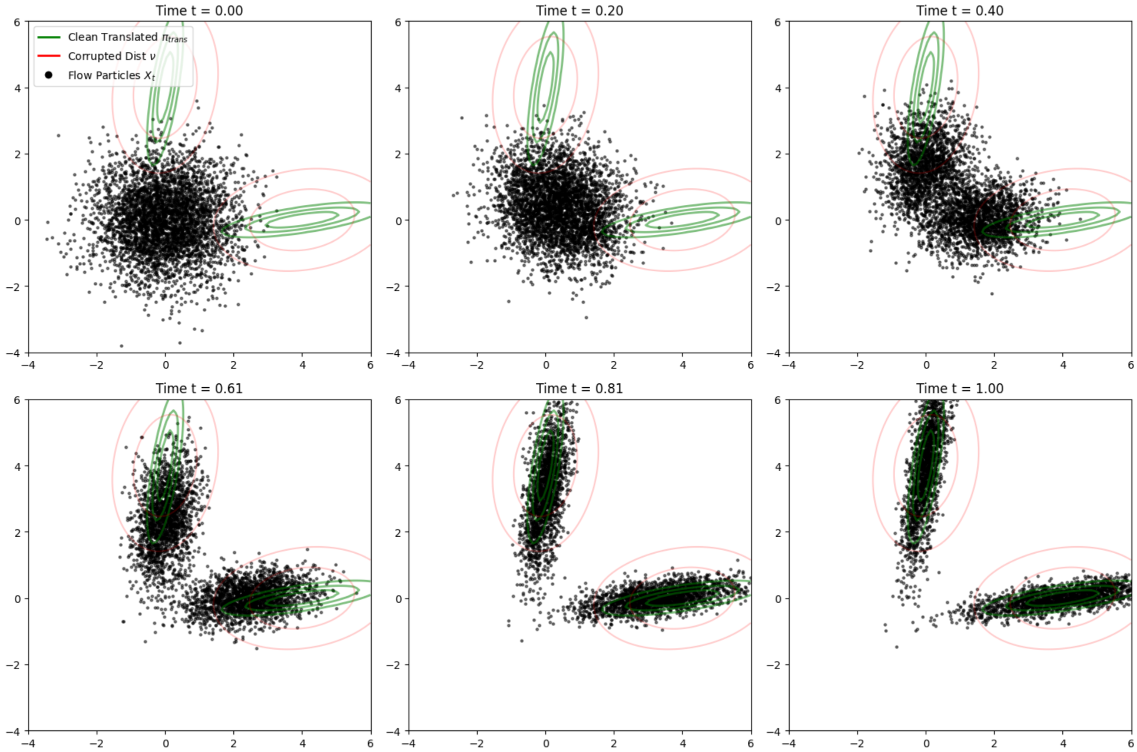

Figure 2: Empirical samples time evolved according to $d\hat{X}_t = b_t(\hat{X}_t|y)\,dt$. At $t = 0$, the samples are sampled from $X_{t=0} \sim \mathcal{N}(0, \mathbb{I})$; each sample gets its own conditioner sampled from $y \sim \nu$ (corrupted distribution).

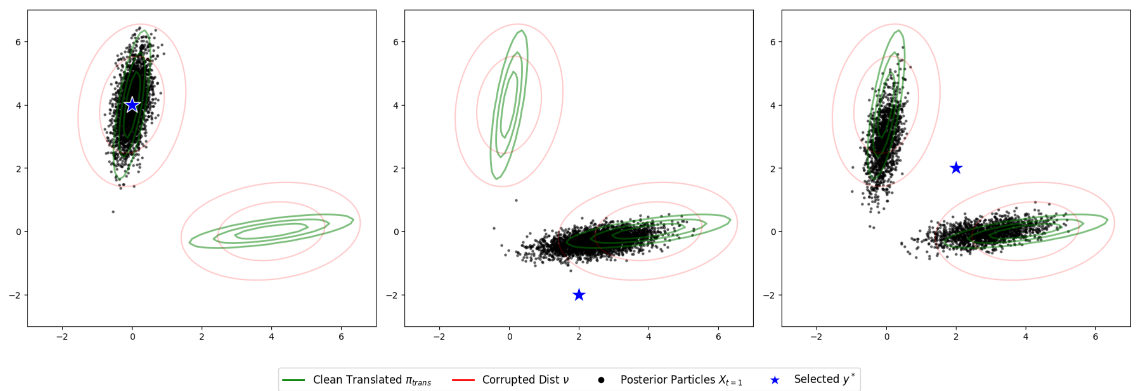

Figure 3: Various outcomes of conditioning on observation y (for $b_t(x|y)$).